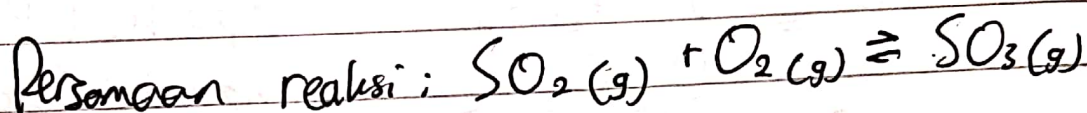


Keseimbangan Kimia

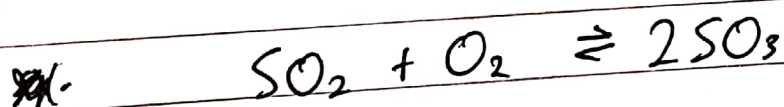
Contoh Soal

- Pada ruang tertutup yang bervolume 5 liter, di reaksi 0,6 mol gas SO_2 dan 0,1 mol gas O_2 . Setelah reaksi terbentuk 0,4 mol SO_3



Tentukan:

- keseimbangan kimia dan keseimbangan tekanan
- K_c
- K_p jika tekanan total 2,2 atm

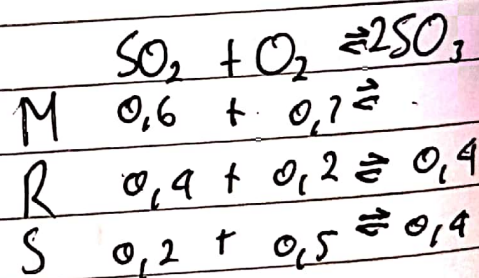


$$K_c = \frac{[\text{SO}_3]^2}{[\text{SO}_2]^2 [\text{O}_2]}$$

$$K_c = \frac{[0,08]^2}{[0,04]^2 [0,1]}$$

$$K_c = \frac{[64/10000]}{[16/10000] [1/10]}$$

$$K_c = 40$$



$$0,2/5\text{L} = 0,04\text{M}$$

$$0,5/5\text{L} = 0,1\text{M}$$

$$0,4/5\text{L} = 0,08\text{M}$$

$$P_{\text{partial}} = \frac{n_{\text{senyawa}}}{n_{\text{total}}} \times P_{\text{total}}$$

$$n_{\text{SO}_2} = 0,2 \text{ mol} \quad n_{\text{O}_2} = 0,5 \text{ mol} \quad n_{\text{SO}_3} = 0,4 \text{ mol}$$

$$n_{\text{total}} = 1,1 \text{ mol}$$

$$P_{\text{total}} = 2,2 \text{ atm}$$

$$P_{\text{SO}_2} = \frac{0,2}{1,1} \times 2,2 = 0,4 \text{ atm}$$

$$P_{\text{O}_2} = \frac{0,5}{1,1} \times 2,2 = 1 \text{ atm}$$

$$P_{\text{SO}_3} = \frac{0,4}{1,1} \times 2,2 = 0,8 \text{ atm}$$

$$K_p = \frac{[P_{\text{SO}_3}]^2}{[P_{\text{SO}_2}]^2 [P_{\text{O}_2}]} = \frac{[0,8]^2}{[0,4]^2 [1]} = 4 \text{ atm}$$

Hubungan K_c dan K_p dan $Suhu$

Rumus

$$K_p = K_c (R \cdot T)^{\Delta n}$$

K_p = Kesetimbangan Tekanan

K_c = Kesetimbangan Kimia

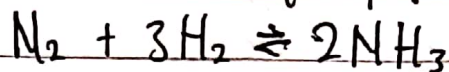
R = Tetapan Gas Ideal ($0,082 \text{ L} \cdot \text{atm} / \text{mol} \cdot \text{K}$)

T = Suhu (Kelvin)

Δn = Selisih koefisien Produk dikurangi Reaktan ($k_p - k_r$)

Contoh Soal

Dalam suatu wadah terdapat kesetimbangan



Nilai K_c pada suhu 500°C adalah 0,04. Hitunglah K_p

Dik:

$$R = 0,082$$

$$\Delta n = k_{\text{ek}} - k_{\text{ri}} = 2 - (1+3) = -2$$

$$T = 500^\circ\text{C} = 773 \text{ K}$$

$$K_c = 0,04$$

$$K_p = K_c (R \cdot T)^{\Delta n}$$

$$K_p = 0,04 (0,082 \times 773)^{-2}$$

$$K_p = \frac{0,04}{(0,082 \times 773)^2}$$

$$K_p = \frac{0,04}{(63,386)^2}$$

$$K_p = 9,9557 \times 10^{-6} \text{ atm}$$